Hackathon Challenge: Build an Impact Metric (IM) for cricketers

Problem Statement (abstract idea)

In cricket, average and strike rate indicate how much and how fast a player scores. However, the sport lacks a unified metric that quantifies how much a player truly influences the outcome of a match.

Your challenge is to design an Impact Metric (IM) that measures the influence a player creates in every match.

Impact is not just about raw performance (runs, wickets, eco etc.)

The most meaningful impact often emerges under pressure.

Impact = Performance × Match Context × Game Situation

Where:

- Performance = measurable output (runs, wickets, economy, etc.)
- Match Context = state of the game (wickets fallen, required run rate, phase of innings, opposition quality, etc.)
- Situation = pressure intensity, match importance and previous out of form phase

The final impact metric needs to be aggregated on a rolling last 10 innings of individual players and visualized on a meter like given below:

Metric Requirements:

- Normalized on a 0–100 scale
- 50 represents a neutral baseline
- Calculated on a rolling last 10 innings
- Updated after every match (not real-time)
- Recency must influence the score

⇒ **You may use the above base framework — but innovation is strongly encouraged.**

Objective

This is not about creating a rule-based scoring system.

It is about designing a:

- Robust
- Data-driven
- Context-aware
- Scalable
- Non-gameable metric

that meaningfully represents player impact.

Expected Deliverables

1. Definition of Impact
   Clearly define what “Impact” means and how pressure, performance, and context contribute to it.

2. Mathematical Formulation
   Present the equation/model behind the Impact Metric (IM), including normalization (0–100 scale) and baseline logic.

3. Model / Algorithm Explanation
   Explain how match context, situation, and recency (rolling last 10 innings) are incorporated.

4. Assumptions & Design Choices
   Key assumptions, weightage logic, and constraints considered.

5. Sample Outputs
   Demonstrate IM calculation using sample players or match scenarios.

6. Robustness
   Show how the metric handles edge cases and avoids artificial inflation.

7. Bonus: Visualization
   Impact meter and trend view across recent innings.

End Goal

When the game gets tough… how impactful is this player?

We are not looking for rule implementation.
We are looking for cricket intelligence translated into data science.